SIMATS ENGINEERING
ASSESSMENT TOOL 2: Scenario-Based

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

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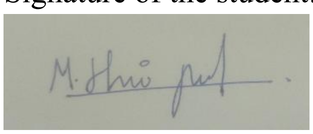
COURSE OUTCOME COVERED
CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Dave's Taxonomy: Manipulation (Level 2) **Total Marks: 25**
Question Count: 5

Code of Conduct
I Hariprasath(192421246) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the student:



Assessment Tasks
Scenario-Based Questions

1. A media company wants to migrate its video streaming platform to the cloud. The system must handle millions of concurrent users with minimal buffering. Design a cloud architecture using scalability and caching techniques. Explain how load balancing and auto-scaling can ensure smooth streaming. Discuss cost optimization strategies for such high-demand workloads.

2. A fintech startup is deploying applications using containers across multiple cloud providers. They face challenges in maintaining consistency and managing deployments. Propose a solution using container orchestration and multi-cloud strategies. Explain how service discovery and scaling can be handled efficiently. Highlight the benefits and risks of a multi-cloud deployment model.

3. An e-learning platform experiences data loss due to accidental deletion in the cloud. The organization needs a robust backup and disaster recovery strategy. Design a solution using cloud-native backup, replication, and versioning. Explain how recovery time objective (RTO) and recovery point objective (RPO) are achieved. Provide steps to ensure business continuity during failures.

4. A government agency is adopting a hybrid cloud model for sensitive and non-sensitive data. Some applications must remain on-premises due to compliance requirements. Design a hybrid architecture that ensures secure communication between environments. Explain how identity management and encryption should be implemented. Discuss challenges in managing hybrid cloud environments.

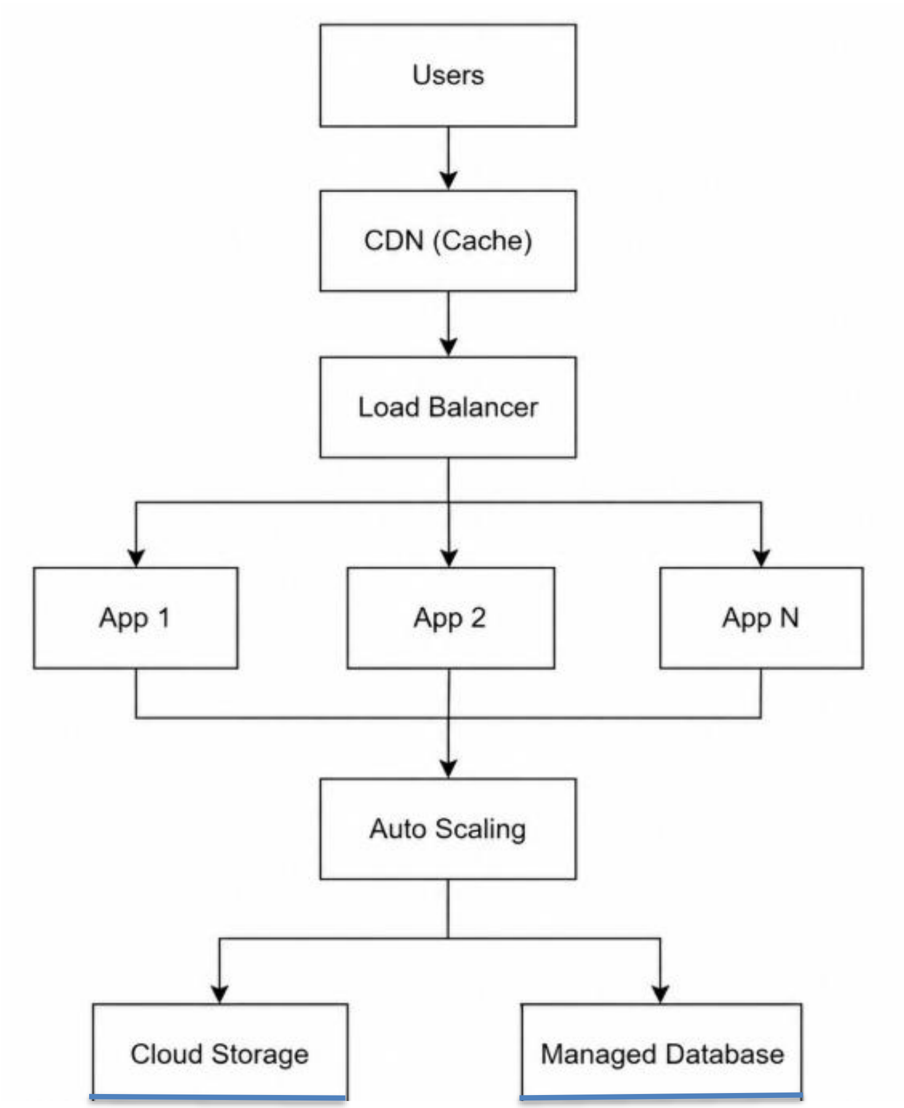
5. A SaaS provider needs to ensure high availability and fault tolerance for its application. The system must continue operating even if one region goes down. Propose a multi-region deployment strategy in the cloud. Explain how data replication and failover mechanisms should be configured. Discuss monitoring and alerting strategies for proactive issue detection.

Scenario-Based Rubric

Criteria	**Weightage**	**Level 4 - Exemplary**	**Level 3 - Proficient**	**Level 2 - Developing**	**Level 1 - Beginning**
Understanding of Scenario	20%	Demonstrates complete understanding of the scenario context	Good understanding with minor gaps	Partial understanding	Fails to understand the scenario
Identification of Risks / Issues	20%	Identifies all major issues correctly	Identifies most issues	Limited issue identification	Misses major issues
Application of Concepts	25%	Applies virtualization concepts effectively	Applies concepts with minor mistakes	Partially correct application	Weak application
Decision Justification	20%	Decisions are well justified and technically strong	Reasonable justification	Weak justification	No useful justification
Clarity	15%	Clear and well-organized response	Generally clear	Somewhat unclear	Poorly presented

1.A media company wants to migrate its video streaming platform to the cloud. The system must handle millions of concurrent users with minimal buffering. Design a cloud architecture using scalability and caching techniques. Explain how load balancing and auto-scaling can ensure smooth streaming. Discuss cost optimization strategies for such high-demand workloads.

(a)Cloud Architecture using Scalability and Caching



(b)Load Balancing and Auto-Scaling

- **Load Balancer** distributes user requests among multiple servers, preventing overload.
- If one server fails, traffic is redirected to healthy servers.
- **Auto Scaling** increases the number of instances during peak hours and decreases them during low traffic.
- This ensures low buffering, high availability, and better user experience.

(c)Cost Optimization Strategies

- Use **Auto Scaling** to avoid paying for idle servers.
- Use CDN caching to reduce bandwidth costs.
- Store frequently accessed videos in high-speed storage and older videos in cheaper archive storage.
- Use Reserved or Spot Instances for predictable workloads.
- Monitor resource usage and remove unused resources.

2.A fintech startup is deploying applications using containers across multiple cloud providers. They face challenges in maintaining consistency and managing deployments. Propose a solution using container orchestration and multi-cloud strategies. Explain how service discovery and scaling can be handled efficiently. Highlight the benefits and risks of a multi-cloud deployment model.

(a) Propose a solution using container orchestration and multi-cloud strategies. Package applications using **Docker containers**.

Solution

- Package applications using **Docker containers** to ensure consistency across environments.
- Use **Kubernetes** for container orchestration to automate deployment, scaling, and management.
- Deploy Kubernetes clusters across multiple cloud providers (AWS, Azure, GCP).
- Store container images in a **Container Registry** for easy access.
- Implement a **CI/CD pipeline** to automate application build, testing, and deployment.
- Use **Infrastructure as Code (IaC)** tools (e.g., Terraform) to provision the same infrastructure across all cloud providers

(b) Explain how service discovery and scaling can be handled efficiently.

Service Discovery

- Kubernetes automatically assigns DNS names to services.
- Containers communicate using service names instead of IP addresses.
- Automatically updates service endpoints when containers are added or removed.
- Ensures seamless communication between microservices.

Scaling

- Use Horizontal Pod Autoscaler (HPA) to increase or decrease pods based on CPU or memory usage.
- Load balancers distribute traffic evenly across available pods.
- Kubernetes automatically replaces failed containers using self-healing.
- Ensures high performance during changing workloads.

(c) Highlight the benefits and risks of a multi-cloud deployment model.

Benefits

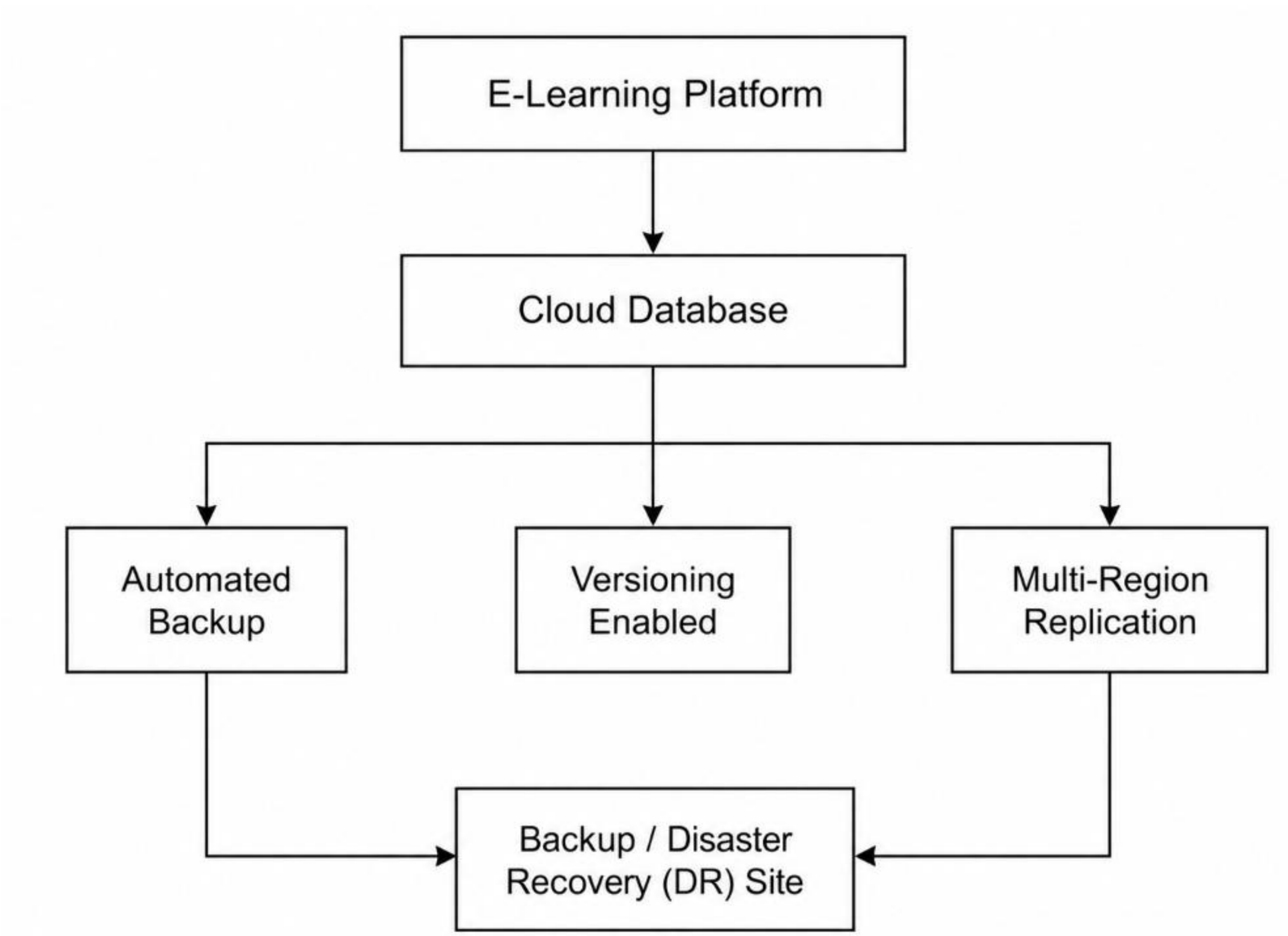
- High Availability: Applications continue running even if one cloud provider fails.
- Avoids Vendor Lock-in: Organizations are not dependent on a single cloud provider.
- Better Disaster Recovery: Data and applications can be recovered from another cloud.
- Improved Performance: Users can access services from the nearest cloud provider.
- Flexibility: Choose the best services and pricing from different cloud providers.

Risks

- Complex Management: Managing multiple cloud environments is more difficult.
- Higher Operational Cost: Additional networking and management expenses.
- Security Challenges: Maintaining consistent security policies across providers.
- Data Synchronization Issues: Keeping data consistent between multiple clouds.
- Monitoring Complexity: Requires centralized monitoring and management tools.

3.An e-learning platform experiences data loss due to accidental deletion in the cloud. The organization needs a robust backup and disaster recovery strategy. Design a solution using cloud-native backup, replication, and versioning. Explain how recovery time objective (RTO) and recovery point objective (RPO) are achieved. Provide steps to ensure business continuity during failures.

(a) Design a solution using cloud-native backup, replication, and versioning.



(b) Explain how Recovery Time Objective (RTO) and Recovery Point Objective (RPO) are achieved.

Recovery Time Objective (RTO)

Definition:
The maximum acceptable time to restore services after a failure.

How it is achieved:

- Maintain a **Disaster Recovery (DR) site** with standby resources.
- Use **automatic failover** to switch to the backup system.
- Restore data quickly using cloud snapshots and backups.
- Regularly test disaster recovery procedures.

Recovery Point Objective (RPO)

Definition:
The maximum acceptable amount of data loss after a failure.

How it is achieved:

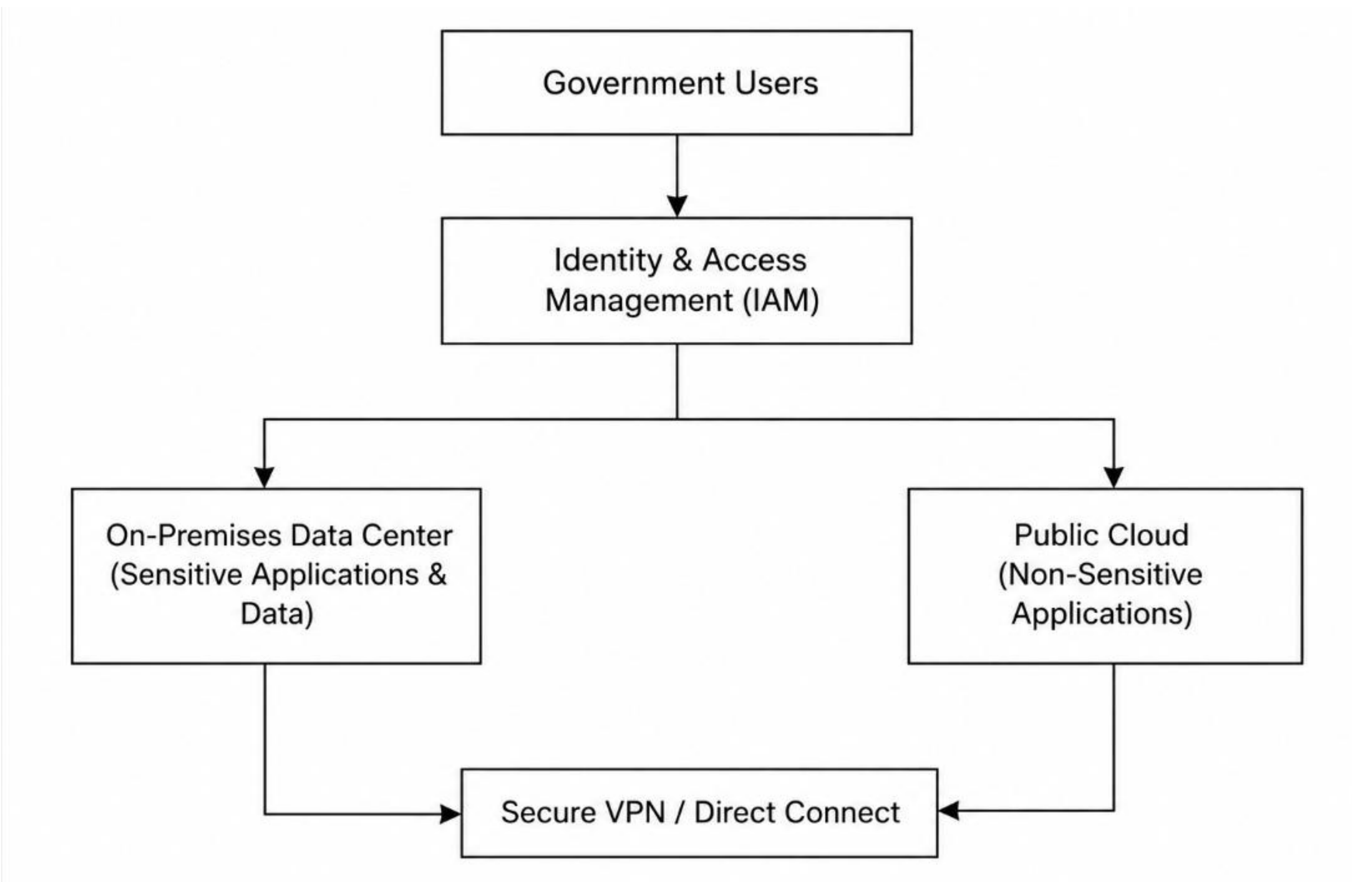
- Perform **frequent automated backups**.
- Enable **real-time data replication** between regions.
- Use **versioning** to recover deleted or modified files.
- Schedule regular database snapshots.

(c) Provide steps to ensure business continuity during failures.

1. Enable **automatic cloud backups** for all critical data.
2. Replicate data to **multiple cloud regions**.
3. Enable **versioning** for cloud storage.
4. Maintain a **Disaster Recovery (DR) site**.
5. Configure **automatic failover** to the backup environment.
6. Monitor backup and replication status continuously.
7. Test backup restoration and disaster recovery plans regularly.
8. Maintain a documented **Business Continuity Plan (BCP)** and train staff to follow recovery procedures.

4.A government agency is adopting a hybrid cloud model for sensitive and non-sensitive data. Some applications must remain on-premises due to compliance requirements. Design a hybrid architecture that ensures secure communication between environments. Explain how identity management and encryption should be implemented. Discuss challenges in managing hybrid cloud environments.

(a) Design a hybrid architecture that ensures secure communication between environments.



(b) Explain how identity management and encryption should be implemented.

Identity Management

- Implement Identity and Access Management (IAM) to control user access.
- Use Role-Based Access Control (RBAC) to assign permissions based on user roles.
- Enable Multi-Factor Authentication (MFA) for secure user authentication.
- Apply the Principle of Least Privilege (PoLP) so users have only the minimum required permissions.

- Monitor user activities through audit logs.

Encryption

- Encrypt data at rest using AES-256 encryption.
- Encrypt data in transit using SSL/TLS protocols.
- Store and manage encryption keys securely using a Key Management Service (KMS).
- Regularly rotate encryption keys to improve security.

(c) Discuss challenges in managing hybrid cloud environments.

- Complex Infrastructure Management: Managing both on-premises and cloud resources increases administrative complexity.
- Security Management: Maintaining consistent security policies across different environments is challenging.
- Compliance Requirements: Government regulations require strict control over sensitive data.
- Network Latency: Communication between on-premises systems and the cloud may introduce delays.
- Integration Issues: Integrating legacy systems with cloud services can be difficult.

5.A SaaS provider needs to ensure high availability and fault tolerance for its application. The system must continue operating even if one region goes down. Propose a multi-region deployment strategy in the cloud. Explain how data replication and failover mechanisms should be configured. Discuss monitoring and alerting strategies for proactive issue detection.

(a) Propose a multi-region deployment strategy in the cloud.

Deployment Strategy

- Deploy the application in multiple cloud regions to eliminate a single point of failure.
- Use a Global Load Balancer to route user requests to the nearest healthy region.
- Deploy application servers and databases in each region.
- Configure Active-Active or Active-Passive deployment for continuous service availability.
- Ensure each region has sufficient resources to handle user traffic if another region fails.

(b) Explain how data replication and failover mechanisms should be configured.

Data Replication

- Enable real-time database replication between all cloud regions.
- Replicate application data and storage continuously to keep data synchronized.
- Maintain backup copies in secondary regions for disaster recovery.

Failover Mechanism

- Configure automatic failover to redirect traffic if the primary region becomes unavailable.
- Use health checks to detect failures in servers or regions.
- Automatically switch users to the healthy region with minimal downtime.
- Regularly test failover procedures to ensure reliable disaster recovery.

(c) Discuss monitoring and alerting strategies for proactive issue detection.

Monitoring

- Monitor CPU, memory, storage, and network utilization.
- Track application response time and server health.
- Monitor database replication status and backup operations.
- Use centralized logging to analyze application and infrastructure events.

Alerting

- Configure alerts for server failures, high CPU usage, network latency, and storage issues.
- Send notifications through email, SMS, or messaging platforms to administrators.
- Generate alerts for replication failures, backup failures, and security incidents.
- Use dashboards to continuously monitor the health of all cloud regions.